

## Performance Testing

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 2 July 2026                     |
| <b>Team ID</b>       | xxxxxx                          |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                                                         |
|----------------------------|---------------------------------------------------------------------------------|
| <b>Testing Tool Used</b>   | Apache JMeter 5.6.3                                                             |
| <b>Type of Testing</b>     | Load Testing                                                                    |
| <b>Target Module / API</b> | Home Page (HTTP GET Request) of Credit Card Approval Prediction Web Application |
| <b>Test Environment</b>    | Local System (Windows 10, Flask Application, Python 3.13)                       |
| <b>Test Date</b>           | 03 July 2026                                                                    |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description  | No. of Virtual Users | Duration (sec) | Expected Outcome                                        |
|------|------------------------------|----------------------|----------------|---------------------------------------------------------|
| 1    | Load Home Page               | 20                   | 10             | Page should load successfully without errors            |
| 2    | Multiple Concurrent Requests | 20                   | 10             | All requests should be processed successfully           |
| 3    | Response Time Measurement    | 20                   | 10             | Average response time should remain low                 |
| 4    | Throughput Evaluation        | 20                   | 10             | Application should handle multiple requests efficiently |

### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value       | Status (Pass / Fail) | Remarks                                     |
|------|----------------------|--------------|--------------------|----------------------|---------------------------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 5 ms               | Pass                 | Very fast response time                     |
| 2    | Response Time (Max)  | < 5 seconds  | 40 ms              | Pass                 | Maximum response time well within the limit |
| 3    | Throughput (Req/sec) | > 10 Req/sec | 10.6 Req/sec       | Pass                 | Stable request handling                     |
| 4    | Error Rate           | < 1%         | 0.00%              | Pass                 | No request failures                         |
| 5    | CPU Utilization      | < 80%        | Below 80% (Normal) | Pass                 | System remained stable during testing       |
| 6    | Memory Utilization   | < 80%        | Below 80% (Normal) | Pass                 | No excessive memory usage observed          |

### Step 4: Observations & Analysis

#### Key Findings

The Credit Card Approval Prediction System was evaluated using Apache JMeter 5.6.3 under load testing conditions. The application successfully processed 100 HTTP requests generated by 20 virtual users. The average response time was 5 milliseconds, while the maximum response time recorded was 40 milliseconds. The system achieved a throughput of 10.6 requests per second with an error rate of 0.00%, indicating reliable and efficient performance.

#### Bottlenecks Identified

No significant performance bottlenecks were observed during testing. The application maintained consistent response times throughout the test, and all requests were completed successfully without failures. The measured response times remained well within the expected limits.

#### Optimization Steps Taken

The application performance was improved through effective data preprocessing, feature engineering, optimized machine learning models, and a lightweight Flask web application. These optimizations enabled fast prediction responses and stable application performance during load testing.

### Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Aggregate Report (mx) (C:\Users\my-pc\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Aggregate Report (mx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:09 0 0/20

Test Plan  
Thread Group  
HTTP Request  
Summary Report  
View Results Tree  
Aggregate Report

Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

| Label        | # Samples | Average | Median | 90% Line | 95% Line | 99% Line | Min | Maximum | Error % | Throughput | Received KB/sec | Sent KB/sec |
|--------------|-----------|---------|--------|----------|----------|----------|-----|---------|---------|------------|-----------------|-------------|
| HTTP Request | 100       | 5       | 5      | 7        | 8        | 9        | 3   | 40      | 0.00%   | 10.6/sec   | 53.80           | 1.83        |
| TOTAL        | 100       | 5       | 5      | 7        | 8        | 9        | 3   | 40      | 0.00%   | 10.6/sec   | 53.80           | 1.83        |

Credit Card Approval Prediction x +

Ask Gemini

127.0.0.1:5000

## Credit Card Approval Prediction

Applicant ID

Gender

Owned Car

Owned Realty

Total Children

Total Income

Income Type

Education

Family Status

Housing Type

Mobile Phone

Work Phone

Phone

Email